

## TECHNICAL COMMUNICATION · FRAMEWORK PROPOSAL

# The PAEV Framework

## A Collaborative Approach for Technical Alignment and Evidence-Based Communication

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| 1.0     | May 2026 | Framework proposal | CC BY 4.0 |

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**Abstract**

Technical environments are increasingly driven by data, analytics, and cross-functional collaboration. Yet, despite considerable technological investment, organizations continue to face recurring failures that originate not from a lack of information, but from misalignment in interpretation. Metric discrepancies, conflicting business definitions, inconsistent requirements, and communication gaps between technical and business teams frequently produce rework, delayed decisions, and unnecessary friction. This paper introduces **PAEV**, *Problem, Analysis, Evidence, Validation*, a structured communication framework designed to support collaborative technical alignment. Unlike traditional approaches such as SCQA or SCR, which prioritize persuasion, PAEV proposes a diagnostic process centered on shared understanding, evidence-based reasoning, and explicit validation before action. The framework was developed through practical experience in data engineering, analytics, and technical leadership contexts, particularly in situations involving data discrepancies, metric governance, and stakeholder decision-making.

**Keywords,** technical communication, data engineering, framework, collaborative alignment, evidence-based reasoning, stakeholder communication, metric governance, analytics leadership.

## 1. Introduction

Modern organizations have invested heavily in data platforms, business intelligence systems, and analytical capabilities. Yet, technical teams still spend considerable time resolving conflicts that arise from different interpretations of the same reality. The recurring patterns are familiar to any practitioner in the field:

- different definitions for the same metric across departments;
- divergent results between dashboards and official reports;
- business rules understood differently by stakeholders and engineering teams;
- technical investigations perceived as accusations rather than diagnostics;
- stakeholder resistance driven by communication style rather than technical substance.

In most of these cases, the underlying issue is not technical complexity, it is *alignment*. Traditional approaches to such conversations often focus on proving who is correct. Technical discussions become debates. Investigations become defensive exchanges. Data, instead of serving as an instrument for understanding, is used as ammunition in favor of predetermined positions.

PAEV was developed to address this gap. Its objective is not persuasion. Its objective is collaborative alignment.

## 2. Background and Motivation

Established communication frameworks emphasize narrative structure, explanation, and persuasion. The most widely adopted in consulting and business environments include:

**SCQA** (Situation, Complication, Question, Answer), introduced by Barbara Minto as part of the *Pyramid Principle*, organizes reasoning hierarchically to deliver conclusions with maximum clarity. **SCR** (Situation, Complication, Resolution) is McKinsey's compact adaptation, widely used in executive communication. Both frameworks excel when the desired outcome is persuasion, when the speaker already knows the answer and seeks to communicate it convincingly.

PAEV addresses a fundamentally different problem. In technical environments, the correct answer is often *unknown* at the beginning of the discussion. Business stakeholders may hold rules unknown to technical teams. Technical teams may possess evidence unavailable to stakeholders. The challenge, therefore, is not to convince, it is to construct shared understanding.

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|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>EXISTING FRAMEWORKS</b><br>Optimize for <b>persuasion</b> .<br><i>The speaker has the answer.</i> | <b>PAEV</b><br>Optimizes for <b>alignment</b> .<br><i>The answer is co-constructed.</i> |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|

Figure 1. Conceptual distinction between traditional persuasion frameworks and PAEV.

### 3. The PAEV Framework

PAEV is structured into four sequential stages. Each stage produces a specific output and serves a distinct purpose in the construction of shared understanding.

## P

#### Problem

Define the observed issue objectively. The objective of this stage is to establish a common understanding of reality before introducing explanations, assumptions, or solutions. The statement must be observable, specific, and free of judgment.

##### EXAMPLE

*"A discrepancy was identified between the sales dashboard and the monthly financial report."*

## A

#### Analysis

Describe how the issue was investigated and present the technical interpretation. Analysis demonstrates method, it transforms assumptions into structured reasoning and signals to the audience that the matter has been examined rather than reacted to.

##### EXAMPLE

*"Investigation revealed that the dashboard calculates revenue based on order date, while the financial report uses payment confirmation date."*

## E

#### Evidence

Support findings with measurable information. Evidence reduces subjectivity and shifts the discussion from opinion to diagnosis. Acceptable forms include metrics, logs, queries, reports, and historical records, always presented with context.

##### EXAMPLE

*"October revenue showed a discrepancy of R\$400K, corresponding exactly to orders created in October but paid in November."*

## V

#### Validation

Confirm alignment or request a formal definition. Validation is the distinguishing component of PAEV: instead of imposing a conclusion, the framework returns ownership to the party with the authority to define the rule. It transforms stakeholders from receivers into participants.

##### EXAMPLE

*"For your business context, should revenue recognition follow order date or payment confirmation date?"*

## 4. Application: A Revenue Discrepancy Case

To illustrate the framework in practice, consider a scenario familiar to most data and analytics teams: a divergence between revenue figures reported by the operational dashboard and the official financial statement. The same scenario, communicated with and without PAEV, produces strikingly different outcomes.

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| <p><b>WITHOUT PAEV</b></p> <p><i>"The dashboard is showing a different number from the financial report. Someone did something wrong."</i></p> <p><b>OUTCOME</b></p> <p>Defensive reactions. Blame-seeking. Issue unresolved.</p> | <p><b>WITH PAEV</b></p> <p><b>P · PROBLEM</b></p> <p>A discrepancy was identified between the sales dashboard and the monthly financial report.</p> <p><b>A · ANALYSIS</b></p> <p>The dashboard uses order date; the financial report uses payment confirmation date.</p> <p><b>E · EVIDENCE</b></p> <p>October: R\$2.3M vs R\$1.9M. The R\$400K gap matches orders placed in Oct but paid in Nov.</p> <p><b>V · VALIDATION</b></p> <p>Should revenue be recognized by order date or by payment date?</p> <p><b>OUTCOME</b></p> <p>Shared diagnosis. Business owns the rule. Aligned execution.</p> |
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Figure 2. The same situation, communicated through two different approaches. The structural difference produces a behavioral difference.

In the unstructured version, the conversation collapses into a search for responsibility. No party assumes ownership; the technical team appears accusatory; the business team becomes defensive; and the underlying inconsistency remains unresolved. The same divergence will likely reappear in the following month.

In the PAEV version, the conversation is reframed. The technical team presents a diagnosis, not an accusation. The discrepancy is reduced to a single, answerable business question. Ownership of the decision returns to the party who holds the authority to define the rule. Once the rule is established, downstream systems can be aligned, and the issue does not recur.

## 5. Why PAEV Works

The effectiveness of PAEV is not a matter of clever phrasing. It rests on four behavioral mechanisms that the structure deliberately encourages.

### 5.1 Reduction of defensive reactions

When a conversation opens with an observable fact rather than an interpretation, the counterpart has nothing to defend. There is no implicit accusation, no claim of blame. The neutrality of the Problem stage lowers the cognitive cost of engagement.

## 5.2 Structured investigation

The Analysis stage demonstrates that the technical team has examined the issue rather than merely reacted to it. It positions the team as investigator rather than judge, a posture that tends to elicit cooperation rather than resistance.

## 5.3 Collaborative decision-making

By explicitly acknowledging that technical teams may not possess all business knowledge, the Validation stage creates shared ownership of the outcome. The stakeholder is no longer a passive audience but a co-author of the resolution.

## 5.4 Alignment before action

PAEV prevents premature implementation. By requiring explicit validation before any change is made, the framework reduces the most expensive class of technical errors: those that result from solving the wrong problem correctly.

## 6. Domains of Application

PAEV is domain-agnostic but particularly well suited to environments where technical findings must be reconciled with business definitions. The following are common contexts in which the framework has been applied:

|                             |                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------|
| <b>Data Engineering</b>     | Metric discrepancies, data quality investigations, pipeline validation, governance discussions. |
| <b>Analytics</b>            | KPI definition, dashboard inconsistencies, business interpretation alignment.                   |
| <b>Software Engineering</b> | Requirement clarification, system behavior investigation, integration issues.                   |
| <b>Technical Leadership</b> | Stakeholder communication, cross-functional alignment, decision support.                        |

*Table 1. Domains in which PAEV has demonstrated practical applicability.*

## 7. Limitations

PAEV is not a universal protocol. It assumes that the situation permits structured analysis, evidence gathering, and dialogue, conditions that are not always present. The framework is not recommended for emergency response, where immediate action takes precedence over alignment; for fully deterministic technical problems, where no business interpretation is required; or for pure persuasion scenarios, where the speaker already holds the answer and the goal is to communicate it convincingly. In such cases, frameworks such as SCQA or SCR remain better suited.

## 8. Future Development

Future iterations of PAEV may incorporate decision matrices to support the Validation stage, governance extensions to formalize the output of validated rules, communication templates for common contexts (email, ticketing systems, presentations), and organizational maturity models to assess adoption. Empirical validation through case studies across organizations of different sizes and industries is also a planned direction. The author welcomes collaboration on such studies and on the integration of PAEV into existing technical documentation workflows.

## 9. Conclusion

Technical problems are, more often than not, communication problems wearing technical clothing. Organizations invest in platforms, architectures, dashboards, and analytics systems, but alignment between people remains a human process. PAEV proposes a structured way to transform technical communication into collaborative diagnosis. It replaces the question *who is right?* with the question *what is the correct shared understanding?*, a shift that, in the experience of the author, consistently changes both the tone and the outcome of technical conversations.

Better conversations produce better decisions. PAEV is offered as one practical instrument toward that end.

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## References and Inspirations

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**About the author.** Vitor Hugo Vasconcelos de Lima leads data and analytics teams in Brazil, working at the intersection of data engineering, analytics governance, and technical leadership. The PAEV framework was developed in response to recurring alignment challenges observed in stakeholder-facing technical work.

*Correspondence and collaboration inquiries are welcome.*